

# Competitive Bid Intelligence & Pricing Analytics

Analytics Consulting Project · Client: A Leading Construction & Infrastructure Firm

*The Problem: Bidding on public infrastructure contracts is highly competitive and largely intuition-driven, making it hard to know when a bid is **priced right, too aggressive, or leaving money on the table.***

## What I Built

- 1

Data Cleaning & Imputation

Ingested and cleaned 3,500+ public bid records, using KNN imputation to fill missing project and pricing fields so the dataset was reliable enough to model.
- 2

Competitor Pattern Detection

Applied LOF anomaly detection to flag competitors with systematically unusual bidding behavior, matching 80% of the firm's own competitor watchlist.
- 3

Win-Probability Modeling

Built a logistic regression model estimating competitor-specific win probabilities, factoring in project scale and geographic pricing effects.
- 4

Bid Strategy Recommendations

Translated model outputs into clear go or no-go signals reviewers could use directly when deciding how aggressively to price a bid.

## Impact

|                                                                       |                                                                |
|-----------------------------------------------------------------------|----------------------------------------------------------------|
| <div>\$4M+</div> <div>Potential revenue upside identified</div>       | <div>264</div> <div>Recoverable analyst hours identified</div> |
| <div>3,500+</div> <div>Public bid records cleaned &amp; modeled</div> | <div>80%</div> <div>Match rate vs. known competitor list</div> |
| <div>28</div> <div>Competitors flagged for anomalous bidding</div>    | <div>90 bps</div> <div>Cut in bid-cost exposure</div>          |